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ARTICLE INFORMATION

Article title: A Multi-Source Open Government Dataset for Luanshya Municipal Council Governance, Financial Budgets, and Constituency Development Fund (CDF) Allocations

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Keywords: Open government data; Municipal budgets; Constituency Development Fund (CDF); Data mining; PDF table extraction; Web crawling; Luanshya Municipal Council

Abstract: This data article presents a multi-source open government dataset collected from the official web portal of the Luanshya Municipal Council (luanshyacouncil.gov.zm), located in the Copperbelt Province of Zambia. The dataset spans municipal administrative activities, local financial budgets, civic news releases, and Constituency Development Fund (CDF) allocations for the period 2022–2026. Data collection was carried out with a Python-based web crawler (requests, BeautifulSoup) that traversed the council website for HTML pages and linked PDF documents, followed by PDF parsing with pdfplumber for digital documents and a pytesseract/pdf2image OCR pipeline for scanned documents. Crawl coverage was independently verified by cross-checking the resulting URL list against a second, separately written crawler. The compiled dataset consists of seven pipe-delimited (|) CSV files encompassing 3,097 itemised municipal budget lines, 260 CDF beneficiary records (bursaries and empowerment grants), 60 approved and proposed community infrastructure projects, 13 priority capital-investment project entries from the Luanshya District Integrated Development Plan (IDP 2023–2033), 16 municipal news posts, a 151-record document index, and a 208-record web-crawling coverage-validation log. Fields that could re-identify or facilitate contact with individual beneficiaries (national registration card numbers, phone numbers, dates of birth) were deliberately excluded from the published beneficiary records. This dataset offers reuse potential for researchers, data scientists, policy analysts, and civic-technology developers conducting automated financial auditing, CDF resource-allocation tracking, natural language processing on municipal communications, and regional urban-planning alignment studies.

SPECIFICATIONS TABLE

Subject	<u>Computer Sciences</u>
Specific subject area	Open government data extraction, municipal financial analytics, and public policy data curation
Type of data	Table, Chart, Graph, CSV (Comma-Separated Values) files
Data collection	Data were collected from the Luanshya Municipal Council website (luanshyacouncil.gov.zm) using a custom Python crawler (requests, BeautifulSoup) to harvest HTML pages and linked PDFs. Digital PDFs were parsed with pdfplumber, while scanned PDFs were rendered with pdf2image and processed using pytesseract OCR; unreliable OCR outputs were flagged and excluded from structured extraction. Included documents covered municipal budgets, CDF records, IDPs and official news. Data were normalized by standardizing headers, resolving encoding issues, removing duplicates, normalizing currency to numeric Kwacha values, and saving UTF-8, pipe-delimited CSV files.
Data source location	Institution: Department of Computer Science, University of Zambia, Lusaka, Zambia Data Collection Origin: Luanshya Municipal Council, Luanshya, Copperbelt Province, Republic of Zambia GPS Coordinates: 13.1367° S, 28.4166° E (-13.1367, 28.4166) Primary Source URL: https://www.luanshyacouncil.gov.zm
Data accessibility	Repository name: Kaggle Dataset title: Luanshya Municipal Council Governance and CDF Dataset (2022–2026) Direct URL: https://www.kaggle.com/datasets/team9-unza/luanshya-municipal-council-dataset Instructions: Open the URL, navigate to the Data tab, and download the pipe-delimited () CSV files.
Related research article	N/A

VALUE OF THE DATA

- These data give researchers and policy auditors machine-readable access to 3,097 line-item municipal budget entries, Local Government Equalisation Fund (LGEF) expenditures, and locally generated revenue sources for a Zambian municipal council, a level of financial detail that is otherwise locked inside unstructured PDF budget books.
- The data can be reused for socio-economic analysis of CDF beneficiary distributions (bursaries, skills training, empowerment grants) and community infrastructure investment across wards in the Luanshya and Roan constituencies, supporting equity and resource-distribution studies.
- The data allow urban planners and civic analysts to cross-reference the priority capital projects listed in the Luanshya District Integrated Development Plan (IDP 2023–2033) against the CDF-funded community projects that were actually approved, enabling benchmarking of planned versus realised investment.
- The included document index and coverage-validation log let other researchers audit and extend the crawl itself by assessing what proportion of the council's public document footprint was captured, which document categories still lack a text layer, and where a second, independently built crawler agreed or disagreed with this pipeline's coverage.
- The news-post corpus offers a structured text resource for natural language processing tasks on Zambian municipal communications, such as named-entity recognition of wards and Kwacha amounts, or topic and sentiment analysis of council announcements.

BACKGROUND

The compilation of this dataset was undertaken as part of the CSC 4792 Data Mining and Warehousing group project at the University of Zambia. The project focused on Luanshya Municipal Council within the broader framework of decentralisation and local governance in Zambia, where local authorities are responsible for planning, budgeting, managing public resources, and supporting community-level service delivery under the Local Government Act No. 2 of 2019 and subsequent amendments. The Local Government (Amendment) Act, 2023 and Local Government (Amendment) Act, 2026 provide additional legislative context for local government financing, including the Local Government Equalisation Fund and Constituency Development Fund (CDF).

The dataset was motivated by the requirement to systematically extract and curate administrative and financial information published through the council's public digital resources. Relevant records were distributed across HTML pages, PDF documents, scanned documents, and WordPress-based news publications. A structured data acquisition and transformation approach was therefore used to convert these heterogeneous sources into machine-readable tables. The compilation captured 3,097 budget line items, 60 CDF project records, and 260 beneficiary entries, together with supporting document, geographic, IDP, and news records. A hand-curated seed list and independent crawl were also used as components of the data acquisition and coverage-validation process.

The resulting data article provides the structured records and processing framework generated during the original CSC 4792 project, allowing the extracted source data to be inspected and re-examined without changing the recorded values.

DATA DESCRIPTION

The repository contains raw source materials and structured processed data collected from publicly available Luanshya Municipal Council resources. The repository is organised into data/raw/ and data/processed/. The data/raw/ directory contains the HTML pages and PDF documents retrieved during data collection, while data/processed/ contains seven pipe-delimited (|) CSV files generated from the collected records. The processed files follow the naming convention db-unza26-csc4792-team9-[description].csv and contain 3,805 records across seven data domains. Table 1 summarises the processed files. **Table 1. Structure and contents of the Luanshya Municipal Council data repository.**

File	Records	Variables	Description
db-unza26-csc4792-team9-budget-lines.csv	3,097	6	Municipal budget line entries extracted from council budget and financial documents
db-unza26-csc4792-team9-cdf-beneficiaries.csv	260	8	CDF beneficiary records, including bursary and skills/empowerment support records, with direct personal identifiers excluded.
db-unza26-csc4792-team9-cdf-projects.csv	60	7	Approved and proposed CDF community infrastructure project records.
db-unza26-csc4792-team9-coverage-validation.csv	208	4	URL-level records used to document and compare crawl coverage.
db-unza26-csc4792-team9-document-index.csv	151	11	Master index of 151 retrieved PDF and HTML documents with extraction metadata, page counts, and text-layer status.
db-unza26-csc4792-team9-idp-priority-projects.csv	13	2	Priority capital project rows extracted from the Luanshya District Integrated Development Plan (2023–2033).
db-unza26-csc4792-team9-news-posts.csv	16	8	Council news posts and press releases extracted from news and notices pages.

Budget line data

The file db-unza26-csc4792-team9-budget-lines.csv contains 3,097 rows and six variables. source_document identifies the source budget document; year records the applicable fiscal year; code contains the budget line code where available; description contains the recorded budget item; amount_type identifies whether the amount is proposed, approved, or actual/expenditure; and amount_kwacha contains the monetary value in Zambian Kwacha.

CDF beneficiary data

The file db-unza26-csc4792-team9-cdf-beneficiaries.csv contains 260 rows and eight variables: source_document, list_type, constituency, ward, beneficiary_name, gender, amount_kwacha, and year. list_type identifies the support category, such as bursary or skills/empowerment grant. The monetary field records the corresponding amount in Zambian Kwacha. Direct personal identifiers such as National Registration Card numbers, telephone numbers, and dates of birth are not included in the curated file.

CDF project data

The file db-unza26-csc4792-team9-cdf-projects.csv contains 60 rows and seven variables: document_title, year, ward, project_name, allocated_amount_zmw, status, and source_url. The records contain CDF projects and their recorded ward, year, project description, allocated amount, status, and source location. Project descriptions include infrastructure categories such as schools, health facilities, and water or sanitation projects where these were specified in the source documents.

Coverage-validation data

The file db-unza26-csc4792-team9-coverage-validation.csv contains 208 rows and four variables: url, in_team_pipeline, in_blessing_crawl, and blessing_category. The Boolean fields indicate whether a URL was captured by the project crawler and by the independent comparison crawl. blessing_category records the category assigned during the comparison crawl.

Document index

The file db-unza26-csc4792-team9-document-index.csv contains 151 rows and 11 variables: local_path, url, doc_type, category, title, http_status, bytes, page_count, has_text_layer, table_detected, and notes. The file records the locally stored path, source URL, document type, content category, title, HTTP status, file size, PDF page count where applicable, presence of an extractable text layer, table detection status, and extraction notes.

The 151 indexed records are categorised as 32 miscellaneous records, 26 CDF beneficiary lists, 25 budget/financial documents, 16 news posts, 15 CDF project lists, 12 notices/minutes, 9 legal/regulatory documents, 8 application forms, 7 navigation pages, and 1 Integrated Development Plan document.

IDP priority project data

The file db-unza26-csc4792-team9-idp-priority-projects.csv contains 13 rows and two variables: page and row_data. page records the page number in the source IDP document,

while row_data contains the cleaned and flattened text of the corresponding extracted table row.

News-post data

The file db-unza26-csc4792-team9-news-posts.csv contains 16 rows and eight variables: source_document, url, title, published_date, wards_mentioned, kwacha_amounts_mentioned, word_count, and body_text. These fields record the source document, publication URL, headline, publication date, wards named in the post, Kwacha amounts mentioned, word count, and the full extracted body text.

All seven processed files use UTF-8 encoding and a pipe (|) delimiter. The processed records were standardised from the collected HTML and PDF sources, with duplicate records from multiple crawls reconciled and differences in source table structures normalised. Missing or incomplete fields were checked against available council publications and independent crawl records.

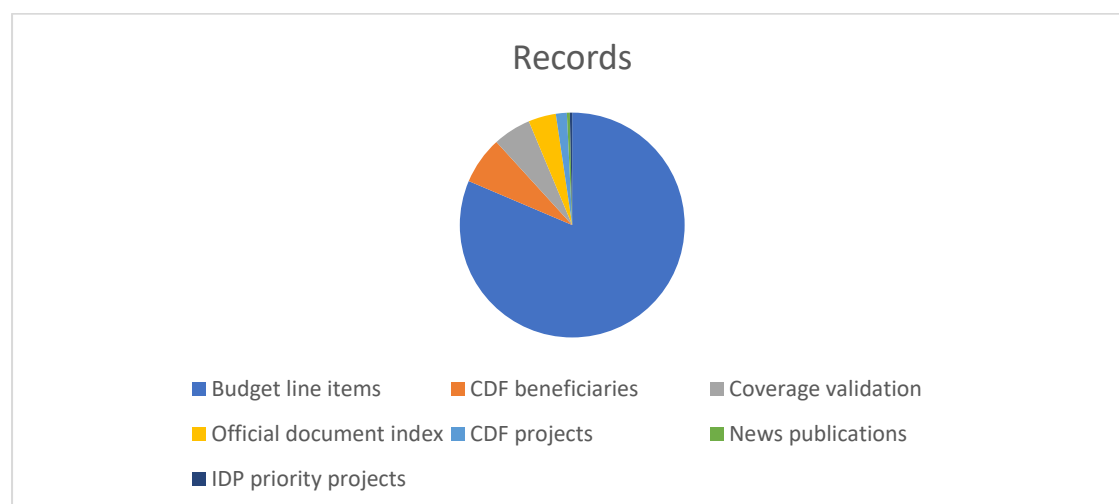


Figure 1. Distribution of records across the seven processed CSV files in the Luanshya Municipal Council repository.

EXPERIMENTAL DESIGN, MATERIALS AND METHODS

The dataset was constructed through a five-stage computational data acquisition and processing pipeline designed to retrieve, inspect, extract, validate, and structure publicly available information from the Luanshya Municipal Council website (luanshyacouncil.gov.zm). The pipeline comprised: (1) web crawling and document harvesting, (2) raw-source storage, (3) document metadata inspection and categorisation, (4) content extraction from HTML and PDF sources, including OCR processing for scanned documents, and (5) data cleaning, normalization, validation, and export to the seven processed CSV files described in Section 2.

The overall processing sequence was:

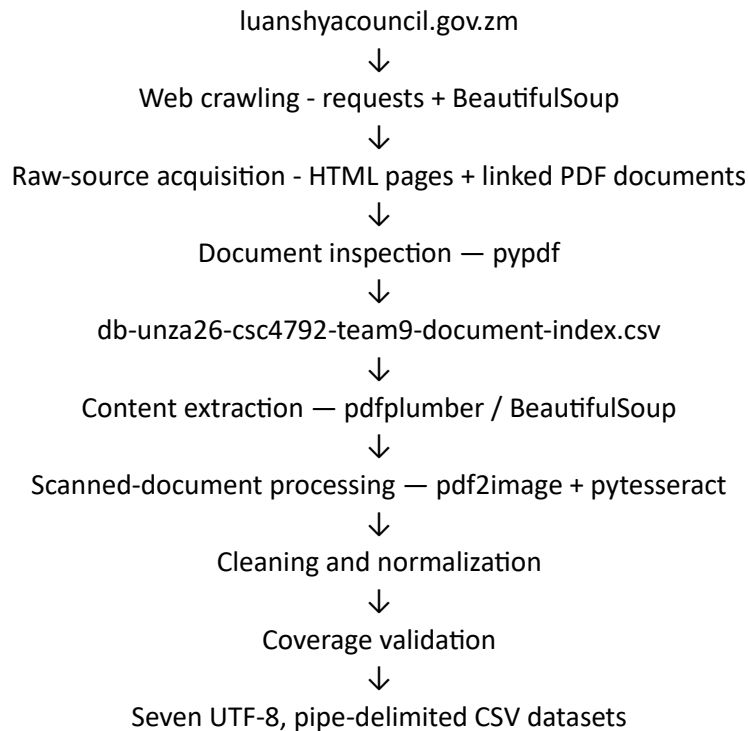


Figure 2. Data acquisition and processing pipeline used to construct the Luanshya Municipal Council dataset.

Web crawling and source identification

Data acquisition targeted publicly accessible digital resources published through the Luanshya Municipal Council website. The source material comprised HTML pages and linked PDF documents containing municipal budgets and financial records, Constituency Development Fund (CDF) project and beneficiary records, Integrated Development Plan (IDP) material, council notices and minutes, legal or regulatory documents, application forms, and institutional news publications.

A hand-curated seed list, stored as `sources_seed.csv`, was used to define the initial URLs for the crawl. The crawler subsequently followed relevant internal links discovered within the retrieved pages. This approach combined predefined source locations with links identified during the crawl.

The crawler was implemented in Python using the `requests` library for HTTP requests and `BeautifulSoup` for HTML parsing. Retrieved HTML pages were examined for hyperlinks and document references. Links pointing to PDF resources were identified and the corresponding documents were downloaded for subsequent processing.

The crawler operated on publicly accessible resources and did not require authentication or access to restricted council systems. During the acquisition period, the municipal domain presented an expired SSL/TLS certificate. Certificate verification was therefore bypassed specifically for requests to `luanshyacouncil.gov.zm` to permit retrieval of the publicly accessible resources. This setting was not applied as a general certificate-verification bypass for unrelated domains.

Execution and acquisition information was recorded in `manifest.csv`, providing a processing record associated with the collected source material.

Raw-source acquisition and organisation

The acquisition stage generated two principal source types: HTML pages and PDF documents. HTML pages were retained as retrieved web resources, while linked PDF documents were downloaded and stored locally.

The raw source material was organised under the repository's `data/raw/` directory. The raw files were retained separately from the processed data so that subsequent extraction and transformation operated on locally stored source material rather than directly on changing web pages.

The repository therefore separates the source and structured layers:

- `data/raw/` — acquired HTML pages and downloaded PDF documents;
- `data/processed/` — structured, normalized CSV datasets.

The raw files provided the input collection for document inspection, classification, and subsequent extraction.

Document inspection and indexing

Following acquisition, the downloaded documents were inspected to determine their file properties and extraction characteristics. The Python `pypdf` library was used to inspect PDF properties including page count, file size, and the presence or absence of an extractable text layer.

The inspection stage produced the master document index:

`db-unza26-csc4792-team9-document-index.csv`

The document index contains 151 records and 11 variables:

- `local_path` — local path to the stored source file;
- `url` — source URL;
- `doc_type` — document type, such as PDF or HTML;
- `category` — assigned content category;
- `title` — document or page title;
- `http_status` — HTTP response status;
- `bytes` — file size in bytes;
- `page_count` — number of pages for PDF documents;
- `has_text_layer` — whether an extractable text layer was identified;
- `table_detected` — whether a table was detected during PDF processing;
- `notes` — document-level processing or extraction notes.

The `has_text_layer` field was used to distinguish machine-readable PDFs from scanned documents requiring image-based processing. The `table_detected` field records whether table structures were identified during PDF extraction.

Source-document categorisation

The indexed documents were assigned categories based on their content and source location. Categories included CDF project lists, CDF beneficiary lists, budget/financial documents, news, IDP material, legal/regulatory documents, notices/minutes, application forms, navigation pages, and miscellaneous records.

The 151 indexed records comprise 32 miscellaneous records, 26 CDF beneficiary lists, 25 budget/financial documents, 16 news posts, 15 CDF project lists, 12 notices/minutes, 9 legal/regulatory documents, 8 application forms, 7 navigation pages, and 1 IDP document.

The category field was subsequently used to identify source documents relevant to the individual structured datasets. Budget and financial documents were processed for budget-line records; CDF beneficiary lists were processed for beneficiary records; CDF project lists were processed for project records; IDP material was processed for priority-project records; and news pages were processed for news-post records.

Extraction from machine-readable PDF documents

PDFs containing extractable text were processed using pdfplumber. The extraction procedure retrieved textual content and identified tabular structures within the documents.

Because the structure and column headings of council documents varied between documents and publication years, keyword-based header matching was used during table extraction. This allowed equivalent fields to be identified when source documents used different column titles or layouts.

Extracted table rows were retained together with source-document identifiers where available. The extracted records were then mapped to the target schemas described in Section 2.

For budget documents, extracted information was transformed into the following fields:

source_document, year, code, description, amount_type, and amount_kwacha.

For CDF project documents, extracted information was transformed into:

document_title, year, ward, project_name, allocated_amount_zmw, status, and source_url.

For CDF beneficiary documents, extracted information was transformed into:

source_document, list_type, constituency, ward, beneficiary_name, gender, amount_kwacha, and year.

Processing of scanned PDF documents

During document inspection, 62 of the 151 indexed documents were identified as scanned documents without a digital text layer. These documents were not processed through text extraction alone.

For image-based processing, PDF pages were rendered to images using pdf2image. The resulting images were passed to pytesseract for optical character recognition (OCR). The OCR process generated machine-readable text from scanned page images.

OCR output was inspected before structured extraction. Where OCR introduced structural or text-recognition errors that prevented reliable tabular extraction, the source document was flagged in the document index and excluded from structured CSV export rather than being force-parsed. The processing status of these documents was retained through the has_text_layer and notes fields of the document index.

This procedure distinguished documents containing directly extractable machine-readable text from documents for which structured extraction depended on OCR.

HTML extraction

HTML-based news and publication pages were processed separately from PDF documents using BeautifulSoup. Relevant page elements were identified from the HTML structure and transformed into the fields defined for the news dataset.

The resulting file,

db-unza26-csc4792-team9-news-posts.csv,

contains eight variables:

- source_document;
- url;
- title;
- published_date;
- wards_mentioned;
- kwacha_amounts_mentioned;
- word_count; and
- body_text.

The source URL was retained for each extracted news record. The body_text field contains the extracted text of the corresponding publication, while wards_mentioned and kwacha_amounts_mentioned record the respective values identified within the post.

Transformation and normalization

Following extraction, records were transformed into the target schemas for the seven processed datasets. Column names were standardized within each dataset, and source records with different table structures were mapped to the corresponding target fields.

Text data were standardized to UTF-8 encoding. The final files were written as CSV files using a pipe (|) delimiter. Monetary fields were converted to numeric Zambian Kwacha values where the source provided an identifiable financial amount.

Embedded line breaks and other formatting artefacts in extracted text were cleaned during transformation. Missing or unavailable fields were retained as missing rather than populated with values not present in the source material.

Source-document identifiers and URLs were retained where available to maintain the association between structured records and their source documents.

Duplicate reconciliation and cross-source checking

Records obtained through multiple acquisition paths were compared during the cleaning stage. Duplicate records arising from repeated crawling or multiple source paths were merged or reconciled where they represented the same underlying source record.

Differences in table layouts between documents were mapped to the corresponding standardized fields. Missing or incomplete values were checked against available council publications and independently collected crawl records where applicable.

The processing did not replace source values with interpreted or estimated values. Where a field could not be established from the available source material, the corresponding value remained missing or was recorded through the applicable processing note.

Personal-data handling

The source material used for the CDF beneficiary records contained personal information. During preparation of the curated beneficiary dataset, direct personal identifiers were excluded from the processed CSV file.

National Registration Card numbers, telephone numbers, and dates of birth were not retained. The processed dataset retained the non-identifying fields specified by its target schema, including constituency, ward, support category, beneficiary name, gender where recorded, year, and financial amount.

Independent crawl and coverage validation

A second crawl was performed independently by a group member to provide a comparison against URLs identified by the primary acquisition process. The comparison records were stored in:

db-unza26-csc4792-team9-coverage-validation.csv

The validation file contains 208 rows and four variables:

- url;
- in_team_pipeline;
- in_blessing_crawl; and
- blessing_category.

The Boolean fields identify whether each URL was captured by the primary project pipeline and the independent comparison crawl. The blessing_category field records the category assigned by the independent crawl.

Of the 208 URLs recorded in the comparison file, 136 were captured by both crawl processes. The remaining URL records represent differences between the two crawl outputs. These differences included URLs associated with WordPress navigation structures, author or category archives, and external links.

The coverage-validation file therefore records URL-level agreement and differences between the two acquisition processes rather than modifying the underlying extracted records.

Generation of processed datasets

After extraction, normalization, duplicate reconciliation, and validation, seven processed CSV files were generated under data/processed/:

1. db-unza26-csc4792-team9-budget-lines.csv — 3,097 records and 6 variables.
2. db-unza26-csc4792-team9-cdf-beneficiaries.csv — 260 records and 8 variables.

3. db-unza26-csc4792-team9-cdf-projects.csv — 60 records and 7 variables.
4. db-unza26-csc4792-team9-coverage-validation.csv — 208 records and 4 variables.
5. db-unza26-csc4792-team9-document-index.csv — 151 records and 11 variables.
6. db-unza26-csc4792-team9-idp-priority-projects.csv — 13 records and 2 variables.
7. db-unza26-csc4792-team9-news-posts.csv — 16 records and 8 variables.

The seven processed files contain 3,805 records in total. The source-document references and URLs retained in the processed datasets provide links between extracted records and the corresponding acquired source material where those fields were available.

Software and computational tools

The complete acquisition and processing workflow was implemented computationally in Python. Table 2 lists the principal software libraries used at each processing stage.

Table 2. Software and tools used for data acquisition and processing.

Software/library	Stage	Function
Python	All stages	Execution environment for acquisition, extraction, transformation, validation, and export
requests	Web crawling	HTTP requests and resource retrieval
BeautifulSoup	HTML processing	HTML parsing, link discovery, and web-page content extraction
pypdf	Document inspection	PDF properties, page counts, and text-layer inspection
pdfplumber	PDF extraction	Text and table extraction from machine-readable PDFs
pdf2image	OCR preparation	Rendering PDF pages as images
pytesseract	OCR	Recognition of text from scanned document images
CSV/Python data-processing functions	Export	Generation of UTF-8, pipe-delimited CSV files

The processing environment did not involve laboratory instruments or physical measurement devices. All source data were obtained digitally and processed using the software components listed in Table 2.

Figure 3. Relationship between acquired source material, document indexing, extraction procedures, validation, and the seven processed datasets.

data/raw/
HTML + PDF source files
↓

pypdf
Document inspection and categorisation
↓
db-unza26-csc4792-team9-document-index.csv
↓
pdfplumber / BeautifulSoup
Text, table and HTML extraction
↓
pdf2image → pytesseract
OCR processing for scanned documents
↓
Cleaning + normalization + duplicate reconciliation
↓
Independent URL comparison
↓
data/processed/
Seven pipe-delimited CSV datasets

LIMITATIONS

Several limitations affected the completeness and standardization of the curated data. Of the 151 indexed source documents, 62 were scanned documents without usable digital text layers. OCR testing with pytesseract was performed, but some outputs were insufficiently reliable for conversion into structured records. Consequently, some beneficiary and financial-statement information could not be included for certain years. In addition, 15 CDF project-list documents required custom table parsing because of merged cells and ward fields that were carried forward across rows. Differences in PDF layouts between annual releases required dynamic, keyword-based header matching, which introduced a risk of column misassignment in non-standard table layouts.

News records were limited to the 16 articles available on the live council website and captured during the crawling period; therefore, the dataset does not represent news content that was unavailable or removed at that time. Differences between the primary and independent crawl inventories also reflected navigation, archive, and external URLs, making source coverage dependent on the accessible website structure.

For privacy protection, National Registration Card numbers, telephone numbers, and dates of birth contained in source beneficiary lists were excluded from the processed data. This reduces the availability of direct identifying fields but limits the level of personal-detail representation in the curated beneficiary records.

ETHICS STATEMENT

This work uses publicly accessible government information published through the official Luanshya Municipal Council web portal (luanshyacouncil.gov.zm). No human-subject research procedures, animal experiments, or data collected from social media platforms were undertaken.

Although the source materials include beneficiary information published by the council, direct personal identifiers were not retained in the curated dataset. National Registration Card numbers, telephone numbers, and dates of birth present in source beneficiary lists were excluded prior to dissemination. The retained fields were limited to information required for the data compilation, including constituency, ward, beneficiary name, gender where reported, year, list type, and amount.

The dataset therefore consists of records derived from publicly accessible municipal publications, with direct personal identifiers removed during data curation.

CRedit AUTHOR STATEMENT

Phil Samakayi: Investigation, Data Curation.

Blessing Yabe: Investigation, Data Curation, Validation.

Lubasi Sipangule: Data Curation, Writing – Original Draft.

Maximillan Soko: Data Curation, Resources, Visualization.

Elijah Zulu: Software, Methodology.

Lighton Phiri: Conceptualization, Supervision, Project Administration.

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DECLARATION OF COMPETING INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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